

MIAE 221 · Materials Science for Engineers

Master Step-by-Step Quantitative Problem Guide

Concordia University · Dept. of Mechanical, Industrial & Aerospace Engineering · Universal Expansion Standard

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Standard: Universal Step-by-Step Expansion Method (Practice Problems, Crystallography & Midterm Prep)

The Step-by-Step Expansion Methodology

Following the universal standard established in `exact_ode_step_by_step.pdf` :

1. **Explicit Step Breakdown:** Every phase is numbered (Step 1 , Step 2 , Step 3 ...).
2. **Theory Before Calculation:** The governing physical law, formula, or geometric theorem is defined before substituting values.
3. **Zero Skipped Calculations:** All unit conversions, exponents, and intermediate products are fully displayed.
4. **"Pattern to Remember" Algorithm:** Every problem ends with an algorithmic checklist to reproduce during exams.

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Problem 1: Percent Ionic Character Calculation (Pauling Formula)

Problem Statement

Calculate the percent ionic character and percent covalent character of the chemical bond in Titanium Dioxide (TiO_2).

Given Pauling electronegativity values: $X_{\text{Ti}} = 1.5$; $X_{\text{O}} = 3.5$.

Step 1: Identify electronegativities and state Pauling's formula

The degree of ionic character between two bonded elements A and B is quantified by Pauling's empirical equation:

where X_A and X_B are the respective Pauling electronegativities.

Step 2: Compute the absolute electronegativity difference ΔX

$$\Delta X = |X_O - X_{Ti}| = |3.5 - 1.5| = 2.0$$

Step 3: Compute the squared difference and the exponent

$$(\Delta X)^2 = (2.0)^2 = 4.0$$

$$\text{Exponent} = -0.25 \times (\Delta X)^2 = -0.25 \times 4.0 = -1.0$$

Step 4: Evaluate the exponential term

$$\exp(-1.0) = e^{-1} \approx 0.36788$$

Step 5: Calculate the percent ionic and covalent character

Since total primary bonding character sums to 100:

Conclusion: TiO_2 possesses a mixed bond that is predominantly **ionic**.

✦ Pauling Ionic Character Pattern to Remember

1. Look up electronegativities X_A and X_B .
2. Compute $\Delta X = |X_A - X_B|$.
3. Calculate the exponent $-0.25(\Delta X)^2$.
4. Evaluate .
5. Deduce .

Problem 2: Equilibrium Interionic Spacing and Bonding Energy**Problem Statement**

(Practice Problem Set #1 · Question 3)

For a potassium-chloride ($K^+ - Cl^-$) ion pair, the attractive and repulsive potential energies vary with interionic separation r according to:

$$E_A(r) = -\frac{1.436}{r} \quad (\text{eV}), \quad E_R(r) = \frac{5.8 \times 10^{-6}}{r^9} \quad (\text{eV})$$

where r is expressed in nanometers (nm).

- (a) Determine the equilibrium interionic separation r_0 .
- (b) Determine the equilibrium bonding energy E_0 .

Step 1: State the equilibrium thermodynamic condition

The net potential energy is the sum of attractive and repulsive energies:

$$E_N(r) = E_A(r) + E_R(r) = -1.436r^{-1} + 5.8 \times 10^{-6}r^{-9}$$

At the equilibrium distance r_0 , the net interatomic force is zero ($F_N = -dE_N/dr = 0$). Therefore, the potential energy is at a minimum:

$$\left. \frac{dE_N}{dr} \right|_{r=r_0} = 0$$

Step 2: Differentiate $E_N(r)$ with respect to r

$$\begin{aligned} \frac{dE_N}{dr} &= \frac{d}{dr} [-1.436r^{-1} + 5.8 \times 10^{-6}r^{-9}] \\ &= (-1.436)(-1)r^{-2} + (5.8 \times 10^{-6})(-9)r^{-10} \\ \frac{dE_N}{dr} &= \frac{1.436}{r^2} - \frac{5.22 \times 10^{-5}}{r^{10}} \end{aligned}$$

Step 3: Set the derivative equal to zero at $r = r_0$

$$\frac{1.436}{r_0^2} - \frac{5.22 \times 10^{-5}}{r_0^{10}} = 0 \implies \frac{1.436}{r_0^2} = \frac{5.22 \times 10^{-5}}{r_0^{10}}$$

Step 4: Isolate and solve for r_0

Multiply both sides by r_0^{10} :

$$1.436r_0^8 = 5.22 \times 10^{-5}$$

Divide by 1.436:

$$r_0^8 = \frac{5.22 \times 10^{-5}}{1.436} = 3.6351 \times 10^{-5}$$

Take the 8th root:

$$r_0 = (3.6351 \times 10^{-5})^{1/8} = \mathbf{0.2789 \text{ nm}} \approx \mathbf{0.279 \text{ nm}} \quad (2.79 \text{ \AA})$$

Step 5: Substitute r_0 into $E_N(r)$ to find the bonding energy E_0

$$E_0 = E_N(r_0) = -\frac{1.436}{0.2789} + \frac{5.8 \times 10^{-6}}{(0.2789)^9}$$

- Attractive term: $-\frac{1.436}{0.2789} = -5.1488 \text{ eV}$
- Repulsive term: $\frac{5.8 \times 10^{-6}}{(0.2789)^9} = \frac{5.8 \times 10^{-6}}{1.0138 \times 10^{-5}} = +0.5721 \text{ eV}$ Sum:

$$E_0 = -5.1488 + 0.5721 = \mathbf{-4.577 \text{ eV}} \approx \mathbf{-4.58 \text{ eV}}$$

★ Potential Well Pattern to Remember

1. Formulate $E_N(r) = -Ar^{-1} + Br^{-n}$.

2. Differentiate: $\frac{dE_N}{dr} = Ar^{-2} - nBr^{-(n+1)}$.
3. Set equal to 0 at r_0 and isolate $r_0^{n-1} = \frac{nB}{A}$.
4. Solve for $r_0 = \left(\frac{nB}{A}\right)^{1/(n-1)}$.
5. Plug r_0 back into $E_N(r_0)$ to find bonding energy E_0 .

Problem 3: Coulombic Electrostatic Force Balance Between Ions

Problem Statement

(Practice Problem Set #1 · Question 4)

Calculate the electrostatic attractive force and repulsive force between a K^+ ion ($z_1 = +1$) and an O^{2-} ion ($z_2 = -2$) at an equilibrium spacing of $r = 1.5 \text{ nm}$.

Step 1: State Coulomb's electrostatic force law

$$F_A = \frac{|z_1||z_2|e^2}{4\pi\epsilon_0 r^2}$$

Step 2: List all physical constants and parameters in SI units

- Electronic charge: $e = 1.602 \times 10^{-19} \text{ C}$
- Permittivity of free space: $\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2/(\text{N} \cdot \text{m}^2)$
- Valence magnitudes: $|z_1| = 1, |z_2| = 2 \implies |z_1 z_2| = 2$
- Distance: $r = 1.5 \text{ nm} = 1.5 \times 10^{-9} \text{ m}$

Step 3: Compute numerator and denominator separately

- Numerator:

$$|z_1 z_2|e^2 = 2 \times (1.602 \times 10^{-19})^2 = 2 \times 2.5664 \times 10^{-38} = 5.1328 \times 10^{-38} \text{ C}^2$$

- Denominator:

$$\begin{aligned} 4\pi\epsilon_0 r^2 &= 4\pi \times (8.854 \times 10^{-12}) \times (1.5 \times 10^{-9})^2 \\ &= 1.11265 \times 10^{-10} \times 2.25 \times 10^{-18} = 2.5035 \times 10^{-28} \text{ N} \cdot \text{m}^2 \end{aligned}$$

Step 4: Compute the attractive force magnitude

$$F_A = \frac{5.1328 \times 10^{-38}}{2.5035 \times 10^{-28}} = \mathbf{2.05 \times 10^{-10} \text{ N}} \quad (\mathbf{0.205 \text{ nN}})$$

Step 5: Apply the equilibrium force balance condition

At the equilibrium distance r_0 , the net force must be zero:

$$F_N = F_A + F_R = 0 \implies F_R = -F_A$$

Since attraction pulls inward ($F_A = 2.05 \times 10^{-10} \text{ N}$), repulsion pushes outward with equal magnitude:

$$F_R = +\mathbf{2.05 \times 10^{-10} \text{ N}} \quad (\mathbf{+0.205 \text{ nN}})$$

★ Coulomb Force Pattern to Remember

1. Convert distance r from nm to meters ($1 \text{ nm} = 10^{-9} \text{ m}$).
2. Use $F_A = \frac{|z_1 z_2| e^2}{4\pi\epsilon_0 r^2}$.
3. At equilibrium, $F_R = -F_A$ (equal magnitude, opposite sign).

Problem 4: Wire Geometry to Microscopic Atom Count Stoichiometry

Problem Statement

(Practice Problem Set #1 · Question 5a)

Calculate the total number of gold (Au) atoms in a cylindrical gold wire with diameter $d = 0.70 \text{ mm}$ and length $L = 8.0 \text{ cm}$.

Given: Density of gold $\rho = 19.3 \text{ g/cm}^3$, atomic mass $A_{Au} = 196.97 \text{ g/mol}$, Avogadro's number $N_A = 6.022 \times 10^{23} \text{ atoms/mol}$.

Step 1: Convert all dimensions to centimeters (cm)

- Diameter: $d = 0.70 \text{ mm} = \frac{0.70}{10} \text{ cm} = 0.070 \text{ cm}$
- Radius: $R = \frac{d}{2} = 0.035 \text{ cm}$
- Length: $L = 8.0 \text{ cm}$

Step 2: Compute geometric cross-sectional area and volume

- Area: $A_{cross} = \pi R^2 = \pi (0.035 \text{ cm})^2 = 3.8485 \times 10^{-3} \text{ cm}^2$
- Volume: $V = A_{cross} \times L = (3.8485 \times 10^{-3} \text{ cm}^2) \times 8.0 \text{ cm} = 0.030788 \text{ cm}^3$

Step 3: Compute total mass from density

$$m = \rho \times V = 19.3 \frac{\text{g}}{\text{cm}^3} \times 0.030788 \text{ cm}^3 = 0.59421 \text{ g}$$

Step 4: Compute number of moles

$$n_{mol} = \frac{m}{A_{Au}} = \frac{0.59421 \text{ g}}{196.97 \text{ g/mol}} = 3.0167 \times 10^{-3} \text{ mol}$$

Step 5: Compute total number of atoms

$$N = n_{mol} \times N_A = (3.0167 \times 10^{-3} \text{ mol}) \times (6.022 \times 10^{23} \text{ atoms/mol})$$

$$N = 1.82 \times 10^{21} \text{ atoms}$$

★ Macro-to-Micro Pattern to Remember

1. Convert all lengths to cm to match density units (g/cm^3).
2. Compute volume $V = \frac{\pi}{4} d^2 L$.
3. Compute mass $m = \rho V$.
4. Compute atoms $N = \frac{m}{A} \times N_A$.

Problem 5: Face-Centered Cubic (FCC) Lattice Parameter & APF Derivation

Problem Statement

Prove from first geometric principles that the Atomic Packing Factor (APF) of a Face-Centered Cubic (FCC) metallic crystal is $\frac{\pi\sqrt{2}}{6} \approx 0.74$.

Step 1: Count the number of atoms per unit cell (n)

- 8 corner atoms, each shared among 8 adjacent unit cells: $8 \times \frac{1}{8} = 1$ atom.
- 6 face-centered atoms, each shared between 2 adjacent unit cells: $6 \times \frac{1}{2} = 3$ atoms.
- Total atoms per unit cell:

$$n = 1 + 3 = 4 \text{ atoms/cell}$$

Step 2: Identify the close-packed direction and touching geometry

In FCC, atoms touch along the **face diagonals** of the cube (the family $\langle 110 \rangle$). Across any cube face of side length a , the face diagonal contains:

- 1 radius from one corner atom: R
- 1 full diameter from the central face atom: $2R$
- 1 radius from the opposite corner atom: R Total diagonal length:

$$\text{Diagonal} = R + 2R + R = 4R$$

Step 3: Relate lattice parameter a to atomic radius R via the Pythagorean Theorem

For a right triangle on the cube face with legs a and a :

$$a^2 + a^2 = (4R)^2 \implies 2a^2 = 16R^2 \implies a^2 = 8R^2$$

Taking the square root:

$$a = \sqrt{8R} = 2\sqrt{2}R$$

Step 4: Calculate the volume of the unit cell (V_c)

$$V_c = a^3 = (2\sqrt{2}R)^3 = 2^3 \times (\sqrt{2})^3 \times R^3 = 8 \times 2\sqrt{2} \times R^3 = 16\sqrt{2}R^3$$

Step 5: Calculate the total volume of atoms inside the unit cell (V_s)

Assuming hard spheres of radius R :

$$V_s = n \times \left(\frac{4}{3}\pi R^3 \right) = 4 \times \frac{4}{3}\pi R^3 = \frac{16}{3}\pi R^3$$

Step 6: Compute the Atomic Packing Factor (APF)

$$\text{APF} = \frac{V_s}{V_c} = \frac{\frac{16}{3}\pi R^3}{16\sqrt{2}R^3} = \frac{16\pi}{3 \times 16\sqrt{2}} = \frac{\pi}{3\sqrt{2}}$$

Rationalize the denominator: $\text{APF} = \frac{\pi}{3\sqrt{2}} = \frac{\pi\sqrt{2}}{3 \times 2} = \frac{\pi\sqrt{2}}{6} \approx 0.7405 \implies 0.74 \text{ (74\%)} \quad \blacksquare$

Problem 6: Body-Centered Cubic (BCC) Lattice Parameter & APF Derivation

Problem Statement

Prove that for a Body-Centered Cubic (BCC) unit cell, $a = \frac{4R}{\sqrt{3}}$ and $\text{APF} = \frac{\pi\sqrt{3}}{8} \approx 0.68$.

Step 1: Count the number of atoms per unit cell (n)

- 8 corner atoms: $8 \times \frac{1}{8} = 1$ atom.
- 1 central atom inside the body: $1 \times 1 = 1$ atom.
- Total atoms per unit cell:

$$n = 1 + 1 = 2 \text{ atoms/cell}$$

Step 2: Identify the close-packed direction

In BCC, atoms touch along the **body diagonals** of the cube (family $\langle 111 \rangle$). Along the 3D body diagonal:

$$\text{Body Diagonal} = R + 2R + R = 4R$$

Step 3: Relate a to R using 3D Pythagorean Theorem

The body diagonal of a cube with edge a satisfies:

$$a^2 + a^2 + a^2 = (4R)^2 \implies 3a^2 = 16R^2 \implies a\sqrt{3} = 4R$$

$$a = \frac{4R}{\sqrt{3}}$$

Step 4: Calculate unit cell volume (V_c)

$$V_c = a^3 = \left(\frac{4R}{\sqrt{3}}\right)^3 = \frac{64R^3}{3\sqrt{3}}$$

Step 5: Calculate atomic volume (V_s)

$$V_s = n \times \left(\frac{4}{3}\pi R^3\right) = 2 \times \frac{4}{3}\pi R^3 = \frac{8}{3}\pi R^3$$

Step 6: Compute APF

$$\text{APF} = \frac{V_s}{V_c} = \frac{\frac{8}{3}\pi R^3}{\frac{64}{3\sqrt{3}}R^3} = \frac{8\pi}{64} \times \sqrt{3} = \frac{\pi\sqrt{3}}{8} \approx 0.6802 \implies 0.68 \text{ (68\%)} \quad \blacksquare$$

Problem 7: Theoretical Density Computation of FCC Copper

Problem Statement

(Lecture 5 · Example Problem)

Calculate the theoretical density of copper (Cu), given:

- Atomic radius: $R = 0.128 \text{ nm}$
- Crystal structure: Face-Centered Cubic (FCC)
- Atomic mass: $A_{Cu} = 63.55 \text{ g/mol}$

- Avogadro's number: $N_A = 6.022 \times 10^{23}$ atoms/mol

Step 1: Identify parameters from crystal structure

For FCC:

- Number of atoms per unit cell: $n = 4$
- Lattice parameter relation: $a = 2\sqrt{2}R$

Step 2: Calculate lattice parameter a in centimeters (cm)

$$a = 2\sqrt{2}(0.128 \text{ nm}) = 2.8284 \times 0.128 \text{ nm} = 0.36204 \text{ nm}$$

Convert from nm to cm (1 nm = 10^{-7} cm):

$$a = 0.36204 \times 10^{-7} \text{ cm} = 3.6204 \times 10^{-8} \text{ cm}$$

Step 3: Calculate unit cell volume V_c in cm^3

$$V_c = a^3 = (3.6204 \times 10^{-8} \text{ cm})^3 = 4.7454 \times 10^{-23} \text{ cm}^3$$

Step 4: State the theoretical density formula

$$\rho = \frac{n \cdot A}{V_c \cdot N_A}$$

Step 5: Substitute all values and compute ρ

- Mass of unit cell:

$$n \cdot A = 4 \times 63.55 \text{ g/mol} = 254.2 \text{ g/mol}$$

- Denominator ($V_c \cdot N_A$):

$$V_c \cdot N_A = (4.7454 \times 10^{-23} \text{ cm}^3) \times (6.022 \times 10^{23} \text{ mol}^{-1}) = 28.577 \text{ cm}^3/\text{mol}$$

- Density:

$$\rho = \frac{254.2}{28.577} = 8.895 \text{ g/cm}^3 \approx 8.90 \text{ g/cm}^3$$

(This theoretical value closely matches the experimental literature density of Copper: 8.94 g/cm^3 , confirming accuracy).

Problem 8: Crystallographic Direction Miller Indices $[uvw]$

Problem Statement

Determine the Miller indices for the direction vector starting at coordinates $(0, 0, 1)$ and terminating at coordinates $(1, 1, 0)$.

Step 1: Identify coordinates of Tail and Head

- **Tail:** $(x_1, y_1, z_1) = (0, 0, 1)$
- **Head:** $(x_2, y_2, z_2) = (1, 1, 0)$

Step 2: Compute Head minus Tail differences

$$\Delta x = x_2 - x_1 = 1 - 0 = 1$$

$$\Delta y = y_2 - y_1 = 1 - 0 = 1$$

$$\Delta z = z_2 - z_1 = 0 - 1 = -1$$

Step 3: Clear fractions and reduce to smallest integers

The vector components are already integers: $(1, 1, -1)$.

Step 4: Format with square brackets and overbars

Enclose in square brackets with no commas; replace -1 with an overbar $\bar{1}$:

$$[11\bar{1}]$$

Problem 9: Crystallographic Plane Miller Indices (hkl)**Problem Statement**

Determine the Miller indices (hkl) of a crystallographic plane that intersects the x -axis at $x = a$, the y -axis at $y = \frac{2}{3}b$, and is parallel to the z -axis.

Step 1: Origin check

The plane does not pass through the coordinate origin $(0, 0, 0)$. No origin shift is required.

Step 2: Determine axial intercepts in units of a, b, c

- x -intercept: 1
- y -intercept: $\frac{2}{3}$
- z -intercept: ∞ (since the plane is parallel to the z -axis)

Step 3: Take reciprocals of the intercepts

- $h = \frac{1}{1} = 1$
- $k = \frac{1}{2/3} = \frac{3}{2}$
- $l = \frac{1}{\infty} = 0$

Step 4: Clear fractions by multiplying by the lowest common denominator

Multiply all three terms by 2:

- $h = 1 \times 2 = 2$
- $k = \frac{3}{2} \times 2 = 3$
- $l = 0 \times 2 = 0$

Step 5: Enclose in parentheses

$$(230)$$

(The family of all symmetrically equivalent planes in cubic crystals is denoted with braces: $\{230\}$).